



Chapter 5:

Computer Forensics



Outlines

- Introduction
- Evidence collection
- Examinations
- Analysis
- Summary



Introduction

- Digital forensics at the system level involves acquiring and analyzing data.
- This chapter covers *forensically sound data collection* and *analysis techniques*, including *live data acquisition*, which may alter the system.
- With modern encryption tools like BitLocker and advances in memory analysis, live data collection, particularly memory imaging, has become more common.



Introduction

- **Evidence analysis** involves developing accurate theories by taking detailed notes on techniques and data reviewed, emphasizing a sound methodology for examining common artifacts.
- **Timestamps** should not be treated as conclusive evidence without corroboration, as they can be easily manipulated; combining multiple sources can validate their accuracy.
- **Corroborating evidence**, such as matching timestamps in file system metadata or related artifacts, can confirm the reliability of timestamps in investigations.
- This chapter introduces file system and operating system **artifacts**, focusing on Windows and Linux systems, while acknowledging the complexity of other OS and file systems beyond its scope.



Evidence Collection

- Evidence collection is vital to digital forensics, requiring proper handling and a *sound chain of custody* to avoid data corruption.
- Standardized checklists or *step-by-step procedures* can guide less experienced employees in routine data collection tasks.
- Complex collection scenarios, such as **memory imaging** or **live data acquisition**, may require skilled analysts to ensure proper execution.
- Proper documentation of the **process**, **evidence state**, **transfer media**, and **chain of custody** is critical for cases potentially heading to court. Analysts must always follow proper methodologies and document evidence collection thoroughly.



Evidence Collection

- **Data Acquisition:** it involves collecting relevant data, such as live data (memory captures) or forensically sound images or copies.
- The acquisition should minimize changes to the system, with collecting data from a powered-off system being the simplest forensically sound method.
 1. The analyst removes the system's hard drive.
 2. The hard drive is connected to the write-blocker's read-only port.
 3. A target device is connected to the write blocker's read-write port.
 4. The write-blocker's communication port is connected to a computer or standalone device.
 5. The analyst creates an image or copy to the target device.
 6. The original is maintained and documented in a chain-of-custody form.
 7. The image or copy is examined or further copied on the analysis platform in a read-only operation.





Evidence Collection...

- Live data acquisition involves collecting data from a running system. It includes:
 - **Memory** – to locate attacker command activities on a system,
 - **mounted files** – to access files that are protected by the OS registry,
 - **registry hives** – to access the master file table or the metadata file,
 - **unencrypted volumes** - to access the full and mounted volumes, including unallocated spaces,
 - **file systems, security files, or open processes.**



Examination

- The next phase is the examination phase, in which relevant digital evidence is **prepared and extracted** from the collected data.
- It includes accessing the disks, examining the **disk structures and file systems**, and getting the advanced formatting, partitioning, and file structures used.





Examination

- Access the files on disks (**metadata files, master file table, File systems**)
- It helps to see the *file structures, access levels, and who accessed which file* and examine if there are *compressions*.
- Record all data examined for forensic investigation.
- Record the files' *standard information attributes* like MAC(modified, accessed, changed) times.
- Record and examine the *filename attributes*, data attributes, index roots, and allocation attributes.
- Check if there are some **orphan files** (a file whose directory has been deleted and whose MFT record is overwritten or lost.



Analysis

- It mainly depends on the goals of the investigation (investigation of a system for malware, timeline analysis, proof of possession of contraband, or collection and curving of images).
- **Analysis Tools:** there are several tools called digital *forensic suites*.
 - tools for different investigation goals.
 - Perform as many analyses as functions as possible.

Forensic Suite	Company
BlackLight	Blackbag Technologies
EnCase	Guidance Software
FTK	AccessData
Nuix Investigation and Response	Nuix
The SleuthKit and Autopsy	Basis Technology
X-Ways Forensics	X-Ways Software Technologies



Analysis ...

- Some other tools are specialized to **parse** and **analyze** one artifact or a group of artifacts. They are specialized tools for a single artifact or group of artifacts.

Analysis tool	Company	Artifacts analyzed
Event Log Explorer	FSPro Labs	Windows event logs
INDXParse.py	Willi Ballenthin	NTFS master file table
Net Analysis	Digital Detective	Internet browser history
Recon for Mac OSX	Sumuri LLC	Mac OSX forensics
RedLine	FireEye	Live response data collection
Registry Browser	Lock and Code	Windows Registry
SQLite Forensic Toolkit	Sanderson Forensics	SQLite database artifacts
Volatility	Volatile Systems	Memory
Windows Prefetch Parser	TZWorks	Windows Prefetch
dtSearch	dtSearch Corp.	Index searching
Scalpel	Golden G. Richard III	File carving
LiME	504ensics Labs	Linux memory acquisition



Analysis ...

- **Timeline analysis:** is the way used by the analyst to perform analysis in a chronological order.
 - It is commonly used
 - Uses a time stamp with any artifacts that maintain timestamps.
 - Timestamp is derived from the artifact listed in the date/time description field.
 - Examples are log2timeline.

Time	Date/Time Description	Event	Notes
2015-07-01 03:26:33	Registry Key LM	HKEY_USERS\C_ Documents_and_Settings_desktop.user_ntuser.dat\Software\Microsoft\Windows\ShellNoRoam\Bags\30\Shel	desktop.user user session indication
2015-07-01 03:29:23	Last Written	C:\Documents and Settings\desktop.user\Application Data	Likely last time user was logged in
2015-07-01 03:29:26	Created	C:\Windows\svch0st.exe	Backdoor
2015-07-01 16:12:22	Event Log	50 Error The RDP protocol component X.224 detected an error in the protocol stream and has disconnected the client.	RDP activity indicator
2015-07-01 20:45:59	Last Written	C:\Documents and Settings\desktop.user\Start Menu\Programs\Accessories\Command Prompt.lnk	Command prompt used
2015-07-01 20:46:12	URL History	http://fileservr.home.net	
2015-07-01 20:47:11	Registry Key LM	HKEY_USERS\C_ Documents_and_Settings_desktop.user_ntuser.dat\Software\Microsoft\Windows\Shell\Bags\1\Desktop	desktop.user user session indication
2015-07-02 03:09:45	Event Log	50 Error The RDP protocol component X.224 detected an error in the protocol stream and has disconnected the client.	RDP activity indicator
2015-07-02 05:15:35	Event Log	50 Error The RDP protocol component X.224 detected an error in the protocol stream and has disconnected the client.	RDP activity indicator
2015-07-02 12:59:04	Event Log	50 Error The RDP protocol component X.224 detected an error in the protocol stream and has disconnected the client.	RDP activity indicator



Analysis ...

- **File Hashing:** is the process of uniquely identifying a file.
 - Different hashing algorithms are used.
 - A hashed file is compared with other file hashes during the incident response to search for a file.
- **Filtering:** differentiate b/n the noise and the evidence.
 - Used as a form of data reduction in forensic analysis
 - Help to filter the date and time, file name, and path.
- **Data Carving:** is the act of searching for a particular string/bytes within a structure.
 - May apply the string search or its binary pattern while searching.



Analysis ...

- **Memory analysis:** it consists of examining a live memory image and page or swap file.
 - Analyze how the virtual memory manager works.
 - Uses different tools like volatility to determine:
 - Running processes,
 - Content of process memory,
 - Commands executed,
 - Strings (including passwords) cached in plaintext,
 - Malware hooking or injecting other processes, and
 - Nonpersistent malware running in memory.



Summary

- Computer forensics was discussed across disk and file system forensics, operating system forensics, and OS artifact forensics.
- The chapter provides a high-level overview to establish a baseline understanding for analysts and researchers.
- Analysts are encouraged to delve deeper into the discussed artifacts for detailed analysis beyond the chapter's scope.
- Identifying anomalies in user-customizable environments is challenging due to vast datasets and evidence sources.
- Analysts should prioritize artifacts that yield the most results with minimal effort rather than examining every byte or file.
- Additional file systems, artifacts, and OS updates pose unique challenges requiring ongoing research and updated tools.



Excercises

1. Explain the difference between acquiring live data or acquiring a forensic image of a powered-down system. What are the complications and impacts of collecting live data versus creating a forensic image of a powered-down system's data?
2. Imagine an NTFS file system that was formatted with the cluster size specified as 8Kb(8192bytes). If a single-byte file was created, how many null bytes would be written to the sector? What would happen to the additional seven sectors in the cluster, and what might they contain?
3. Explain what a partition is and where hidden data could be placed intentionally by a technically apt individual.
4. On an NTFS file system, there can be more than one attribute of a given type. Explain theoretically where someone can hide data for an executable file in a legitimate system configuration file using the behavior of the filesystem attributes.



Excercises

5. NTFS maintains at least two attributes with MACE(modified, accessed, created, and entry modified) times for every file and directory on the system. Explain the two attributes and describe the differences in how the timestamps on these attributes can be updated.
6. Can deleted files be recovered from EXT file systems? Explain some of the challenges for recovery.
7. Explain why an attacker may use applications, such as WinRAR, to stage data for theft. What artifacts could an examiner look for in memory and virtual memory in order to determine if an attacker used WinRAR and derive the attacker's password?
8. A single date or time on a system can be unreliable, given the many ways that they can be manipulated or even updated through the normal course of the operating system's functions. What type of analysis can be used to corroborate the accuracy of an event's timestamp? What other evidence might be identified from this technique?



Thank you very much!